2022

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-711

[Internet of Things (Elective - I)]

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer ***any five*** questions : 2×5

   (a) What are the differences between 'edge' and 'cloud' computing?

   (b) Justify which one is a preferable option in IoT GLONPAN or GLO.

   (c) Compare CoAP against MQTT.

   (d) What would be a preferable option considering communication – RPi or Aurdino?

   (e) Mention four advantages of using Kafka.

   (f) Why no SQL DB are used in IoT?

2. Answer ***any five*** questions : 4×5

   (a) What are the differences between data analytics and data analysis?

   (b) What is Symbiotic Trust? Explain different aspects of Symbiotic Trust.

   (c) Compare XMPP vs MQTT.

   (d) Compare XMPP vs Web Socket.

   (e) Define SOC. How is it used in IoT development?

   (f) Write a pseudo code for switching an LED from any SOC depending on the input.

3. Answer ***any four*** questions :

   (a) What is Industry 4.0? How does it impact utility and Manufacturing Industry? Explain a typical life cycle of Industry 4.0. 2+2+2+4

   (b) Define four pillars of Industry 4.0. What is MFDM? Why is MFDM preferred over automation? 4+2+4

   (c) Draw a neat diagram to depict different layers of IoT design Methodology. Explain how it can be used for developing a smart university. 5+5

(d) (i) What are the differences between supervised and unsupervised ML? Explain with example.

(ii) Define : Data, Information, Knowledge, and Wisdom.

(iii) Explain how we can achieve wisdom from data with an example of detecting 'Tachycardia'. (2+2)+3+3

(e) Which protocol would you prefer to transmit video data? Explain with reason. Write a process flow chart of an IoT analytics system to make a 'computer vision based occupancy analytics' system. Explain it. 4+6

(f) Explain Moore's law, Kryder's law, and Neilson's law. How they impacted development of IoT? Explain the role of fundamental research behind development of IoT. (2×3)+4